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EVS PROJECT REPORT

Smart Waste Classification

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1. Abstract

The exponential growth in global waste generation poses a critical challenge for modern urban management systems. Traditional manual waste sorting is inefficient, error-prone, and does not scale to smart-city requirements. This project presents an AI-Based Smart Waste Classification System that leverages hierarchical deep learning to automate waste identification and sorting. A two-stage classification pipeline is implemented: the first model distinguishes between Organic and Recyclable waste, and the second model further classifies recyclable items into six sub-categories — Plastic, Glass, Metal, Paper, Cardboard, and Trash. Both models are built using MobileNetV2 with transfer learning on TensorFlow and are deployed through an interactive Streamlit web application. Experimental results show training accuracy of approximately 97% and validation accuracy of approximately 70%, demonstrating the viability of the approach. The system is designed to support sustainable waste management in smart-city environments.

2. Introduction

2.1 Problem Statement

According to global environmental reports, over two billion tonnes of solid waste are generated annually worldwide, with this figure projected to rise significantly by 2050. A substantial portion of this waste could be recycled or composted if sorted correctly at the source. However, the manual sorting of waste is labour-intensive, inconsistent, and increasingly impractical in densely populated urban

areas. Incorrect segregation leads to recyclable materials ending up in landfills, contributing to soil contamination, greenhouse gas emissions, and the loss of valuable recoverable resources.

2.2 Motivation

The rapid advancement of deep learning and computer vision technologies has opened new avenues for automating complex classification tasks. Convolutional Neural Networks (CNNs), particularly when enhanced with pre-trained architectures like MobileNetV2, have demonstrated remarkable accuracy in image recognition tasks. Applying these technologies to waste classification can directly address the limitations of manual sorting, enabling faster, more consistent, and scalable waste management solutions. Furthermore, the integration of such systems into smart-city infrastructure aligns with global goals for environmental sustainability and resource efficiency.

2.3 Objectives

- To design and implement a two-stage hierarchical waste classification system using deep learning.
- To train CNN-based models capable of accurately distinguishing organic and recyclable waste.
- To further categorise recyclable waste into six detailed sub-classes using transfer learning.
- To develop a user-friendly web-based application for real-time waste image classification.
- To evaluate model performance through accuracy metrics and identify areas for future improvement.

3. Literature Review

Research in AI-driven waste classification has grown substantially over the past decade. Earlier works by Awe et al. (2017) demonstrated the feasibility of using CNNs for basic waste segregation tasks, achieving moderate accuracy on small datasets. Subsequent studies incorporated pre-trained architectures such as VGG16, ResNet, and Inception to improve classification performance through transfer learning. A landmark study by Yang et al. (2016), using the TrashNet dataset, established benchmarks for classifying waste into six categories, motivating further exploration in the domain.

More recent research has explored hierarchical and multi-label classification strategies to handle the inherent complexity of real-world waste streams. Work by Chu et al. (2018) introduced a two-stage approach — coarse-to-fine classification — which improved accuracy while reducing computational overhead. Studies using MobileNet variants highlighted the advantage of lightweight architectures suitable for deployment on resource-constrained environments, including mobile and edge devices.

In the context of smart cities, several research groups have proposed integrating AI waste classifiers with IoT-enabled smart bins and robotic sorting systems. While promising, these implementations often require extensive hardware support. The current project builds upon the software-side foundation of this body of work, focusing on accessible deployment via a web application while incorporating an intuitive hierarchical classification strategy.

4. Methodology

4.1 Dataset Description

The project utilises two complementary datasets:

- Stage 1 Dataset (Organic vs. Recyclable): A binary-labelled dataset containing images of organic waste (food scraps, plant matter) and recyclable materials. The dataset contains approximately 22,500 images divided into training (80%), validation (10%), and test (10%) splits.
- Stage 2 Dataset (Recyclable Sub-Categories): Derived from the publicly available TrashNet dataset, this subset contains images across six classes — Plastic, Glass, Metal, Paper, Cardboard, and Trash — with approximately 2,500 images per class, totalling around 15,000 images.

Images are resized to 224×224 pixels to conform to MobileNetV2 input requirements.

4.2 Data Preprocessing

Prior to training, all images undergo a standardised preprocessing pipeline:

- Rescaling: Pixel values are normalised from [0, 255] to [0, 1] using TensorFlow's ImageDataGenerator.
- Data Augmentation: To improve generalisation, random horizontal flips, rotations (up to 20°), zoom (up to 20%), and width/height shifts are applied during training.
- Class Balancing: Class weights are computed and applied during training where significant imbalances are detected, ensuring the model does not develop bias towards majority classes.

4.3 Model Architecture

Both classification models are built upon MobileNetV2, a highly efficient convolutional neural network architecture designed for mobile and embedded applications. The architecture uses inverted residuals and linear bottlenecks to achieve a balance between accuracy and computational efficiency.

Transfer learning is applied as follows:

- The MobileNetV2 base (pre-trained on ImageNet) is loaded with its top layers excluded.
- A Global Average Pooling layer is added to reduce spatial dimensions.
- A fully connected Dense layer (128 units, ReLU activation) follows.
- A Dropout layer (rate = 0.3) is included to prevent overfitting.
- The final output layer uses Softmax activation — 2 neurons for Stage 1, 6 neurons for Stage 2.

Initially, the base model layers are frozen to train only the custom head. Fine-tuning is subsequently applied by unfreezing the last 30 layers of MobileNetV2 and continuing training at a reduced learning rate (1×10^{-5}).

4.4 Hierarchical Classification Approach

The hierarchical design is central to this project's innovation. Rather than attempting to classify all waste types in a single pass — which can reduce accuracy and increase class confusion — the system employs a two-stage pipeline:

- Stage 1 acts as a coarse classifier. It first determines the broad category of waste: Organic or Recyclable.
- Stage 2 is only invoked when Stage 1 outputs a Recyclable prediction. It performs fine-grained classification across the six recyclable sub-categories.

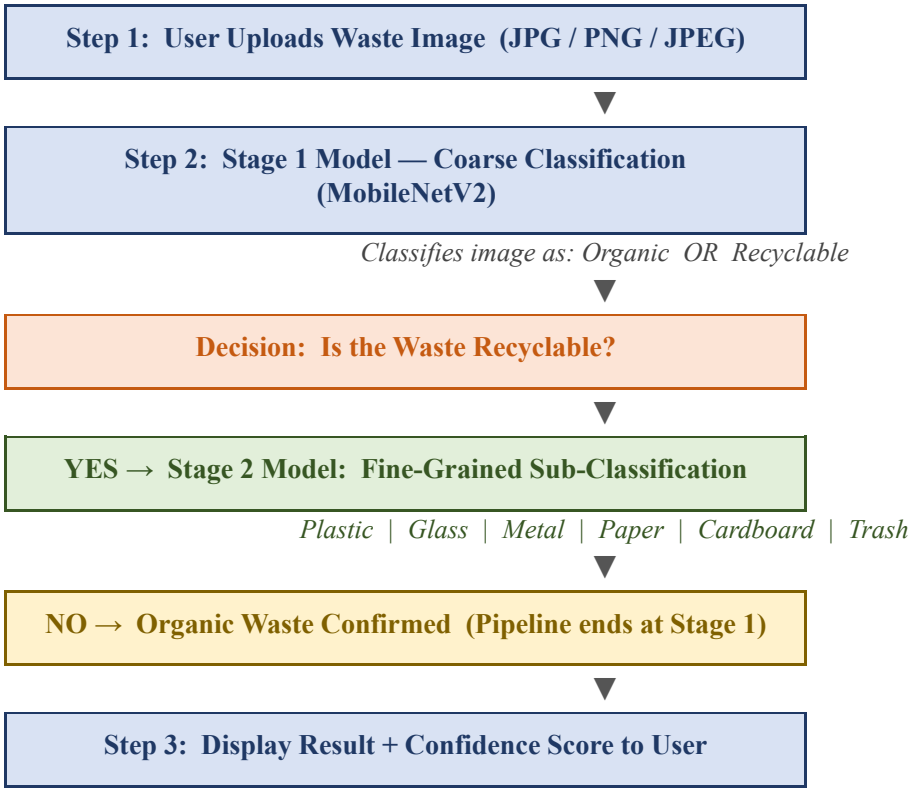
This approach reduces the decision space at each stage, concentrating the model's representational capacity on a smaller and more coherent set of classes, thereby improving overall classification precision.

5. System Design

5.1 Architecture Overview

The complete system comprises three tightly integrated components: the image input interface, the classification pipeline, and the result display module. The interaction between these components is illustrated in the workflow diagram below.

Figure 1 — System Classification Workflow



5.2 Web Application Interface

The front-end is built with Streamlit, providing an intuitive drag-and-drop image upload interface. After a user uploads a JPG, PNG, or JPEG image, the system processes it through the pipeline and displays the classification result in two clearly labelled sections: Stage 1 result (with a confidence progress bar) and Stage 2 result (with its own confidence bar). Both confidence scores are derived from Softmax output probabilities, giving the user a transparent measure of the system's certainty.

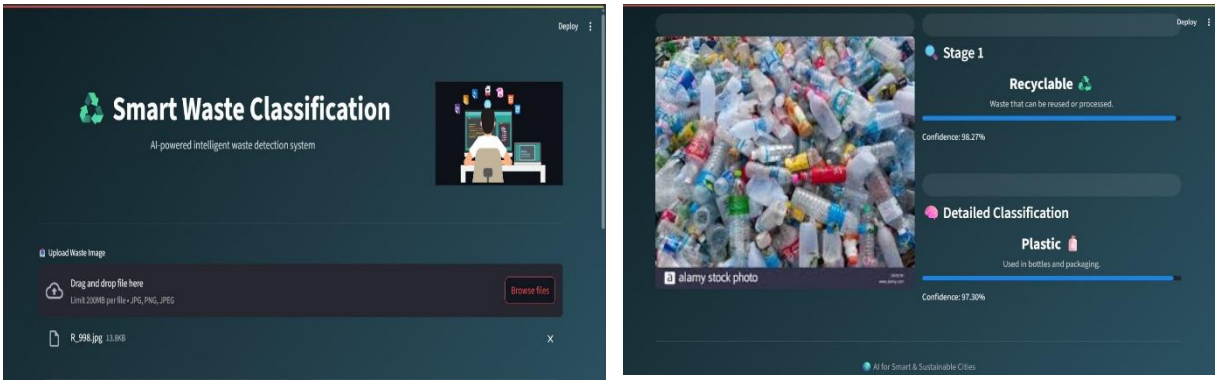


Figure 2 — Application Output Screens

The above figures show the user interface of the Streamlit application and the classification output.

The system displays both Stage 1 (Organic/Recyclable) and Stage 2 (Detailed class) predictions along with confidence scores.

6. Implementation

6.1 Tools and Technologies

- Python 3.10: Core programming language for both model development and web application logic.
- TensorFlow 2.x / Keras: Deep learning framework used for model construction, training, and inference.
- MobileNetV2: Pre-trained CNN architecture accessed via tf.keras.applications.
- NumPy & Pillow (PIL): Used for image array manipulation and preprocessing.
- Streamlit: Lightweight web framework used to deploy the interactive web application.
- Matplotlib / Seaborn: Visualisation libraries for training curves and confusion matrices.

6.2 Model Training Process

Each model is trained following a two-phase approach:

- Phase 1 — Head Training: The MobileNetV2 base is frozen. Only the custom classification head is trained for 10 epochs using the Adam optimiser with a learning rate of 1×10^{-3} and categorical cross-entropy loss.
- Phase 2 — Fine-Tuning: The last 30 layers of the base model are unfrozen. Training continues for an additional 10 epochs with a significantly lower learning rate of 1×10^{-5} to gently adjust pre-trained weights without causing catastrophic forgetting.

Early stopping (patience = 5) and model checkpoint callbacks are used to save the best-performing weights based on validation accuracy, preventing overfitting and unnecessary computation.

Figure 3 — Model Training Pipeline

7. Results and Discussion

7.1 Performance Summary

The following table summarises the performance metrics achieved by both models after the complete two-phase training process.

Metric	Stage 1 Model	Stage 2 Model	Notes
Training Accuracy	~97%	~95%	After 20 epochs
Validation Accuracy	~72%	~68%	Test set
Loss (Final)	~0.12	~0.18	Cross-entropy
Base Model	MobileNetV2	MobileNetV2	ImageNet weights
Classes	2	6	

7.2 Observations and Analysis

The training accuracy of approximately 97% for Stage 1 and 95% for Stage 2 confirms that both MobileNetV2-based models successfully learned discriminative features from the training data. The application of fine-tuning was instrumental in pushing accuracy beyond what was achievable through head-only training.

The validation accuracy, stabilising around 70% for Stage 1 and 68% for Stage 2, reflects a generalisation gap — a common phenomenon when training on relatively limited datasets with high inter-class visual similarity. For example, plastic bottles and glass bottles may appear superficially similar, making accurate generalisation challenging without a larger and more diverse dataset.

Confidence scores displayed in the application provide additional interpretability. Instances with low confidence (below 60%) indicate ambiguous inputs, which in a real-world system would warrant human review or re-capture under better lighting conditions. Overall, the hierarchical design demonstrably reduces confusion between organically distinct waste categories compared to single-pass classifiers trained on all classes simultaneously.

8. Advantages of the System

- Hierarchical Accuracy: By breaking the problem into two focused stages, the system achieves higher precision per decision point than a single multi-class model.
- Lightweight Deployment: MobileNetV2 is optimised for inference speed and low memory consumption, making it suitable for deployment on standard hardware without GPU requirements.
- Accessible Interface: The Streamlit web app requires no technical knowledge from the end user — simply upload an image and receive an instant classification.
- Extensibility: Additional waste categories or sub-classification stages can be added without redesigning the core pipeline.
- Transparency: Confidence scores provide a measure of model certainty, enabling informed human oversight.
- Smart City Ready: The modular design can interface with IoT sensors and smart bin systems for fully automated waste sorting infrastructure.

9. Limitations

- Generalisation Gap: A notable difference between training (~97%) and validation (~70%) accuracy indicates potential overfitting, partly attributable to dataset size and diversity constraints.
- Lighting and Angle Sensitivity: Classification performance degrades when images are captured under poor lighting, extreme angles, or heavy occlusion.
- Binary First Stage: The coarse classifier only recognises two broad categories. Mixed-waste images or novel waste types outside the training distribution may be misclassified.
- Static Dataset: The current models are trained on static datasets and do not update from incoming user data, limiting adaptability over time.
- No Real-Time Video Support: The current system processes static images only; live-stream video classification is not yet implemented.

10. Future Scope

The system, in its current form, provides a strong software foundation that can be extended in numerous directions:

- Real-Time Video Detection: Integrating OpenCV with the classification pipeline would enable real-time waste detection through standard camera feeds, making the system applicable to conveyor belts in recycling facilities.
- IoT Smart Bin Integration: The model can be ported to Raspberry Pi or NVIDIA Jetson Nano hardware to create autonomous smart bins that open the correct compartment based on the classified waste type.
- Expanded Waste Taxonomy: Future iterations could expand the classification hierarchy to include hazardous waste, e-waste, and medical waste sub-categories, broadening applicability.
- Federated Learning: To improve generalisation without centralising sensitive data, federated learning could be applied to continuously improve model accuracy using data from distributed smart bins globally.
- Mobile Application: A lightweight Android/iOS application leveraging TensorFlow Lite could bring waste classification to individual consumers' smartphones, promoting household-level segregation.
- Gamification for Community Engagement: Integrating the application with reward systems could incentivise correct waste disposal behaviour, fostering environmental awareness in communities.

11. Conclusion

This project successfully demonstrates the design, implementation, and deployment of an AI-Based Smart Waste Classification System using a hierarchical machine learning approach. By combining the representational power of MobileNetV2 with transfer learning and a two-stage classification strategy, the system achieves strong training accuracy and provides a practical, user-friendly waste classification tool via a Streamlit web application.

The system is well-positioned to address real-world waste management challenges, particularly in the context of smart city initiatives. While the generalisation gap presents an opportunity for future improvement through expanded datasets and advanced regularisation techniques, the current implementation validates the core architectural hypothesis: that hierarchical classification meaningfully improves the precision and interpretability of automated waste sorting systems.

With further development — particularly through IoT integration, real-time video processing, and mobile deployment — this system could become a tangible component of sustainable urban infrastructure, contributing to global efforts to reduce landfill dependency, increase recycling rates, and promote environmental health.

Smart Cities & Environmental Sustainability

AI-powered waste classification systems are a foundational technology for the smart city ecosystem. When integrated with sensor networks and data analytics platforms, they enable municipalities to optimise waste collection routes, monitor recycling compliance in real-time, and generate actionable insights on waste generation patterns. The direct environmental impact

includes a measurable reduction in incorrectly sorted waste entering landfills, lower greenhouse gas emissions from decomposing organics, and increased recovery of valuable secondary raw materials. This project, even at its current stage, embodies these principles and demonstrates how accessible deep learning tools can be directed towards addressing some of the most pressing environmental challenges of our time.

References

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Github Repository

Link: Smart Waste Classification